



Integration of Nano-Aeration and Microbial Techniques to Control Harmful Algal Blooms and Eutrophication in Reservoirs

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Abstract Eutrophication and harmful algal blooms (HABs) pose a severe threat to freshwater ecosystems, creating an urgent demand for experimentally validated, environmentally sustainable control strategies. Reservoirs are particularly susceptible to eutrophication and HABs due to unique hydrological and ecological traits, including limited water exchange and weak hydrodynamic conditions. Here, we evaluated the efficacy of an integrated environmental-ecological control approach (combining nano-aeration and microbiological techniques) through a one-year in-situ experiment in three eutrophic reservoirs in

Fujian Province, China. We found that the dominant phytoplankton phylum shifted from bloom-forming *Cyanophyta* to *Bacillariophyta* across all reservoirs. Specifically, total phytoplankton density decreased significantly by 87.2%, and the relative abundance of *Cyanophyta* averagely declined by 60.3%, while the relative abundance of *Bacillariophyta* increased by 225.2%. These ecological shifts were closely associated with elevated dissolved oxygen levels and reduced nutrient concentrations (e.g., total phosphorus and total nitrogen) induced by the integrated intervention, especially for the regulation of *Cyanophyta* dynamics. We revealed positive correlations between changes in nutrient concentrations and shifts in *Cyanophyta* relative abundance during the intervention period ($P < 0.05$), emphasizing the necessity of long-term monitoring and adaptive management. This study demonstrates the potential of integrating nano-aeration and microbiological techniques for sustainable eutrophication and HAB's management, and highlights the importance of continuous monitoring to prevent HAB recurrence.

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1 Introduction

Eutrophication is a pervasive issue affecting freshwater worldwide, leading to harmful algal blooms (HABs) that severely degrade water quality (Choudhury et al., 2022; Li et al., 2021; Reid et al., 2019). This phenomenon is primarily driven by excessive nutrient inputs, particularly nitrogen and phosphorus from sources such as agricultural runoff, untreated sewage, and industrial discharges (Malone & Newton, 2020). Recent studies indicate that global climate change and human activities exacerbate the HABs, with increased frequency and intensity of algal blooms being linked to rising temperatures and nutrient runoff (Hu et al., 2024; Woolway et al., 2024). These HABs disrupt aquatic ecosystems and pose significant health risks to local communities by depleting oxygen levels, causing fish kills, and introducing harmful algal toxins (e.g., *Microcystins*, *Cylindrospermopsins*) into water supplies (Kazmi et al., 2022; Ly et al., 2024). Therefore, there is an urgent need to adopt multiple control measures to mitigate the direct consequences of HABs.

Conventional approaches to the management of HABs encompass physical, chemical and biological methodologies (Anderson, 2009). Physical methods (e.g., mechanical removal, light limitation) are localized and ineffective for widespread blooms (Montemazzani et al., 2015), while chemical algaecides (e.g., copper sulfate) risk secondary pollution (Jančula & Maršálek, 2011). Biological controls, such as bio-manipulation with filter-feeding fish, are selective and often insufficient (Anabtawi et al., 2024). Recent studies have emphasized the value of integrated approaches to enhance effectiveness (Frau, 2022). However, there are still significant gaps in the sustainable, large-scale application of these approaches, particularly in reservoirs, where stagnant conditions are conducive to HABs (Kibuye et al., 2021).

The utilization of nanobubble aeration technology has emerged as a promising physical–chemical tool for the management of HABs. The generation of high-pressure shock waves upon the rupture of micro-nano bubbles (50–200 nm diameter) has been demonstrated to result in the damage of Cyanophytal gas vesicles and the subsequent sinking of the organism (Yao et al., 2024). Furthermore, they have been shown to enhance dissolved oxygen (DO) levels, disrupt thermal stratification, and promote

nutrient degradation at the sediment–water interface, which are known to be hostile to *Cyanophyta* (Summers & Ryder, 2023). In shallow lakes, nanobubble aeration has been observed to shift phytoplankton communities from *Cyanophyta*-dominated to *Bacillariophyta*- or *Chlorophyta*-dominated (Chen et al., 2024); however, its efficacy in reservoirs (which possess distinct hydrology) remains understudied (Ali et al., 2023). Bacteria-based biocontrol constitutes an additional sustainable strategy, with particular strains specifically targeting HABs through nutrient competition, antibiotic production, or algal lysis (Pathak et al., 2018). Among the most promising species are *Bacillus subtilis* (suppresses growth of *Cyanophyta* via surfactants (John & Rose, 2005)), *Pseudomonas aeruginosa* (produces algicidal compounds (Morgan-Kiss et al., 2006)), and nitrifying-denitrifying bacteria (reduces nutrient availability (Sun et al., 2018)). It is noteworthy that nanobubble aeration and bacterial biocontrol may synergize: aeration enhances aerobic bacterial activity, while bacteria accelerate nutrient removal, thereby reinforcing aeration-induced conditions. However, the combined use of these substances in reservoirs has not yet been the subject of a systematic evaluation.

In this study, we conducted a one-year in-situ study on three eutrophic reservoirs in Fujian Province, China, to evaluate the efficacy of an innovative integration of nano-aeration and native environmentally friendly microbial techniques. Our objectives were to determine how this synergistic approach regulates nutrient levels and dissolved oxygen concentrations, and reshapes phytoplankton community composition, with a focus on bloom-forming *Cyanophyta*. By disentangling the links between the integrated intervention, environmental drivers, and algal dynamics, we aim to fill the gap in sustainable, reservoir-specific HABs and eutrophication control, and provide mechanistic insights for scalable application.

2 Material and Methods

2.1 Site Selection

We selected three eutrophic reservoirs as our study sites (Fig. 1): Dongyaxi Reservoir (DYX; 118.57°E, 26.55°N), Shanmei Reservoir (SM; 117.85°E, 24.73°N), and Niuyang Reservoir (NY; 117.98°E,

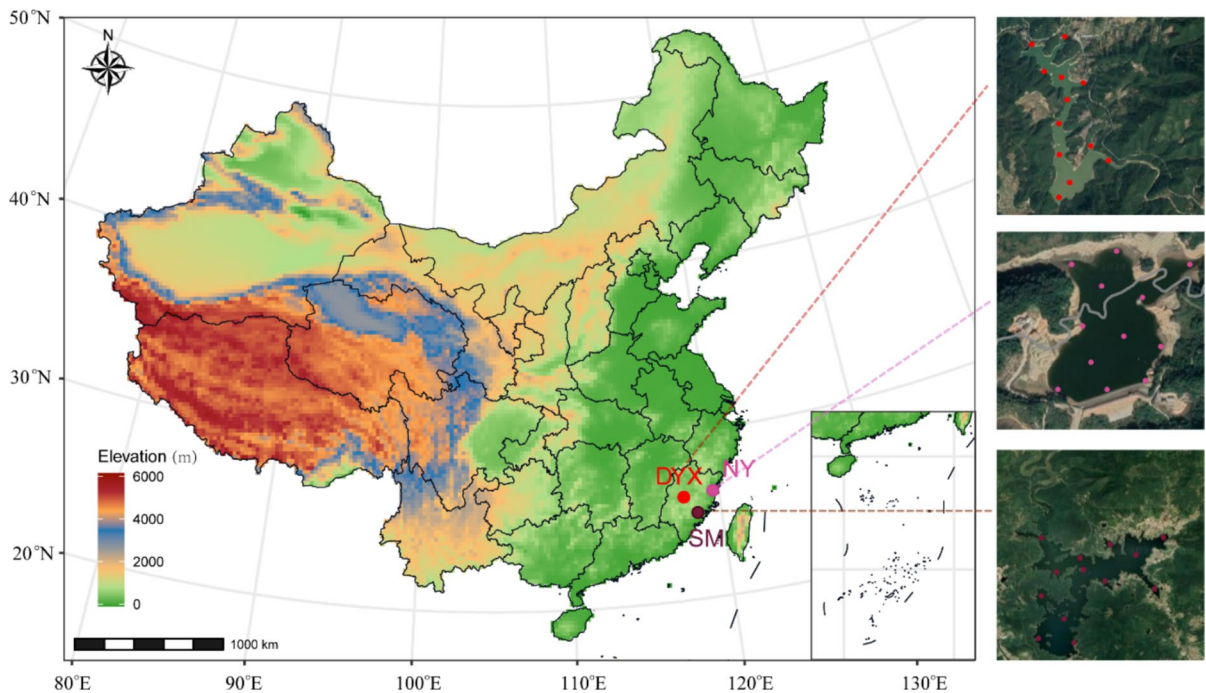


Fig. 1 Locations of the three reservoirs in this study. The colored regions on the map indicate different elevations, with green representing lower elevations and brown/red representing higher elevations. The points with different colors represent the Dongyaxi reservoir (DYX), Shanmei reservoir (SM),

and Niuyang reservoir (NY). Insets on the right display sampling sites of each reservoir. The left map was created using R (Version 4.3.3), and the right figures were sourced from Google Maps

25.05°N) in Fujian Province, China. We chose these three reservoirs for controlling because they have similar conditions of climate (subtropical climate), dominant HABs-forming species (i.e., *Microcystis aeruginosa* and *Anabaena*) and nutrient conditions (Table S2). All the reservoirs have experienced eutrophic conditions every summer, with significant alkaline pollution and algal blooms from June to September, resulting in elevated surface water pH and low dissolved oxygen (DO) levels.

2.2 Experimental Setup

2.2.1 Application of Nano-Aeration Technology

Here, the equipment parameters for our nano-aeration system are listed as follows (Wu et al., 2022; Xiao et al., 2019): **I.** Micro-nano bubble generation device: This device generates micro-nano bubbles with small bubble diameters. It can utilize air, molecular oxygen, or active oxygen (such as hydroxyl radicals) as

gas sources, with oxygen concentrations ranging from 80–100 mg/L. The device is constructed from composite materials and 316L stainless steel, ensuring it is corrosion-resistant, pressure-resistant, and non-contaminating to drinking water. The control system allows for manual or automatic operation, parameter adjustments, data history review, and data export. **II.** Micro-bubble diffusion equipment: This equipment diffuses the micro-nano bubbles produced by the generation device and promotes water flow. It operates at water depths ranging from 0 to 100 m and at pressures greater than 1.0 MPa. Made from stainless steel and composite materials, it is safe for use in drinking water. The diffusion distance of the bubble's ranges from 180 to 9000 m. **III.** Controllable nano oxygenation array disks: These disks disperse gas to form small bubbles, enhancing water circulation. They operate at depths of 0 to 100 m with adjustable array configurations. The bubble diffusion distance ranges from 48 to 2400 m, and each disk services an area of 3600 to 5000 m².

Nano-aeration systems were deployed in a standardized array to ensure uniform treatment across each reservoir. Specifically, in each reservoir:

1. Equipment quantity and placement

12 array-controlled nano-aeration panels (diameter=0.8 m) were installed at 500-m intervals along the reservoir's longitudinal axis, positioned 4–5 m below the water surface to maximize vertical mixing. Four microbubble diffusers (output range=20–30 L/min) were placed at the inflow and outflow zones to enhance horizontal circulation.

2. Operational parameters

The micro-nano bubble generator produced bubbles with a diameter of 50–200 nm, an oxygen transfer efficiency of 35–40% (measured via the Winkler titration method), and a continuous output of 1.2 m³/h of oxygen-enriched gas (85% O₂). Devices operated 12 h/day (08:00–20:00) throughout the 1-year experiment.

3. In-situ monitoring

To verify equipment functionality, dissolved oxygen (DO) was monitored daily at 12 fixed sampling points (consistent with phytoplankton sampling sites) using a calibrated YSI ProPlus probe (precision ±0.1 mg/L). Measurements were taken at 09:00 and 17:00 to capture diurnal variations, and data were logged to confirm DO levels remained within the target range (6–8 mg/L) during operation. In our study, DO monitoring logs indicated >95% operational uptime of the aeration devices, with no significant drops below the target range (8 mg/L) during the experimental period.

2.2.2 Application of Environmentally Friendly Microorganism Technology

Both microorganisms and phytoplankton utilize nutrients in water and have similar environmental requirements (e.g. temperature, dissolved oxygen and pH) (Lage et al., 2021). Here, we isolated beneficial strains of native functional microorganisms from water and sediment samples from the target reservoirs, including acidifying bacteria (*Lactobacillus*

spp.), bacilli (*Bacillus subtilis*), photosynthetic bacteria (*Rhodobacter sphaeroides*), and nitrifying-denitrifying bacteria (*Nitrosomonas europaea* and *Nitrobacter spp.*) (Yang et al., 2011; Huang, 2016).

Firstly, water and sediment samples were collected from multiple locations in the reservoir to ensure a comprehensive representation of the microbial diversity. Subsequent screening included morphological examination and biochemical testing to initially identify strains. For molecular identification, Polymerase Chain Reaction (PCR) amplification of the 16S ribosomal RNA (rRNA) gene was performed using the primers 5'-AGAGTTTGATCCTGGCTCAG and 5'-TACCTTGTTACGACTT. The PCR reaction system (25 µL total volume) contained 12.5 µL of 2×Taq PCR Master Mix, 1 µL of each primer (10 µM), 2 µL of genomic DNA template (50 ng/µL), and 8.5 µL of sterile deionized water. The amplification program was set as follows: initial denaturation at 95 °C for 5 min; 35 cycles of denaturation at 95 °C for 30 s, annealing at 55 °C for 30 s, and extension at 72 °C for 90 s; and a final extension at 72 °C for 10 min. The PCR products were purified and subjected to Sanger dideoxy sequencing, followed by BLAST analysis against the GenBank database to confirm the identity of isolated strains (Basak et al., 2024). These samples were then used to culture the target microorganisms in a controlled environment (e.g., *Lactobacillus spp.* was cultivated at 37 °C in MRS medium (Menon et al., 2013), *Bacillus subtilis* was cultivated at 37 °C in Luria–Bertani medium (Hoover et al., 2010), photosynthetic bacteria was cultivated in the presence of a propionate and light (Takabatake et al., 2004), and Nitrifying-denitrifying bacteria were cultivated in aerobic or anaerobic conditions in specific ammonia-oxidizing or nitrite-oxidizing media (Fu et al., 2022)). After cultivation, the microorganisms were gradually acclimatized to the lower temperatures of the natural reservoirs (20 °C) by exposing them to decreasing temperatures over 7-days period.

Next, a carbohydrate source (e.g., steamed rice) is placed in a small wooden box or perforated plastic container, which is then slightly buried in an area of nutrient-rich sediment. Cover the container with breathable material to ensure air circulation while preventing contamination by larger debris. Depending on environmental conditions, microbial colonies form on the carbohydrate source after an incubation period of 5–10 days (Yassine et al., 2016). The collected

microbial biomass is then mixed with an equal amount of molasses or sugar and fermented for at least one month. This fermentation process increases the microbial population in preparation for its introduction into the water system (Abdel-Rahman et al., 2013). The resulting microbial solution is diluted to a concentration of 0.1–0.2% with unchlorinated water and distributed evenly throughout the reservoir by surface spraying or direct injection into sediment-rich areas (Mao et al., 2021).

2.2.3 Control Design

To isolate the effects of nano-aeration and microbial treatments, we established two types of controls:

1. Spatial controls

Within each reservoir, three 10,000 m² “reference zones” were designated, located > 200 m from treatment zones to avoid aeration diffusion. These zones were not equipped with nano-aeration devices and received no microbial inoculation, but were monitored identically (same sampling frequency and parameters) to treatment zones.

2. Temporal controls

Pre-treatment data (May 2018) from all 36 sampling points served as baseline controls, ensuring comparability with post-treatment data (May 2019) under similar seasonal conditions.

2.3 Data Collection and Analysis

On May 17, 2018, we conducted a survey of the three reservoirs (i.e., DYX, SM and NY) without applying any ecological control techniques and included 12 sampling points for each reservoir. On May 17, 2019, after the application of environmental-ecological restoration for one year, we repeated the data collection at the same 36 points (that is, in total 72 samples). Water and sediment samples were collected from 12 evenly distributed points in each reservoir using a systematic grid approach to ensure spatial representativeness. At each point, 1 L of water was sampled at a depth of 1–2 m, while sediment samples were collected using a grab sampler. Samples were stored on ice during transport to the

laboratory for analysis within 24 h. For phytoplankton collection and identification, we collected 1 L water using a water sampler and fixed with 10 mL of Lugol’s solution (consisting of 20 g potassium iodide, 10 g iodine, and 200 mL glacial acetic acid diluted to 1 L with distilled water) at each sampling site (O’Halloran et al., 2006). The samples were then allowed to settle for 24 h and then concentrated to a volume of 30–50 mL. In the laboratory, a 0.1 mL aliquot was placed in a phytoplankton counting chamber and species identification and cell counting were performed under a microscope (Olympus CX 21) at 400× magnification (Naldi et al., 2024). Identification and quantification (density and biomass calculations) of phytoplankton species were conducted based on relevant literature, with all individuals identified to the genus or species level where possible. We measured several water and sediment parameters, including water dissolved oxygen (DO), water pH, water temperature, water total phosphorus (TP), KMnO₄ index (COD_{Mn}), water total nitrogen (TN), water ammonia nitrogen (NH₄⁺-N), water chlorophyll-a (Chla), sediment TP, sediment TN, and sediment ammonia nitrogen. The DO and pH were measured using a mobile pH probe (Sanxin, China). The NH₄⁺-N was determined in the water using a flow injection analyzer (Skalar, SA1000). The total nitrogen (TN) and total phosphorus (TP) were determined by peroxydisulfate oxidation spectrophotometry, and COD_{Mn} was determined using a Hach HQ40d (Hach, USA) with COD probe. Each parameter was measured four times throughout the day within 6 h, and the mean values was used to represent the environmental parameters for that site.

We first assessed the total density of phytoplankton or the relative abundance of different algal phyla and the correlation between these biological metrics and the measured environmental factors. Spearman’s rank correlation coefficients were calculated with 1000 bootstrap resamples to ensure robust and reliable results, and we visually inspected scatterplots of the data for each pair of variables to confirm that monotonic relationships were present. To quantify the changes in algal-related parameters and environmental factors after the environmental-ecological control, we used the natural log response ratio (lnRR) (Monti et al., 2015). The lnRR was calculated using the following formula:

$$\ln RR = \ln(\bar{X}_t / \bar{X}_c) = \ln \bar{X}_t - \ln \bar{X}_c \quad (1)$$

where $\bar{X}_t$ is the observed values of different parameters after the environmental-ecological control and $\bar{X}_c$ is the observed values of control groups (i.e., before the control).

Next, we used a linear model: $\text{lm}(y \sim 1)$, to evaluate the changes in different parameters for each reservoir, which provided the significant P values (two-ways test) and estimate slopes. For overall assessment, we employed mixed-effects models of:

$$\ln RR = \beta + \pi_{\text{site}} + \epsilon \quad (2)$$

where π_{site} is a random effect explaining the auto-correlation between 3 reservoirs. After obtaining the $\ln RR$ for environmental factors and total phytoplankton density (or the relative abundance of different algal phyla), we conducted correlations to explore the relationships between changes in environmental factors and algal metrics:

$$\ln RR_{\text{alg}} = \beta + \beta_1 \cdot \ln RR_{\text{env}} + \pi_{\text{site}} + \epsilon \quad (3)$$

The model assumptions were validated by checking the normality of residuals (Q-Q plots and histograms) and homogeneity of variances (Levene's test), and significance was determined using post-hoc tests corrected for multiple comparisons with the Benjamini–Hochberg procedure.

Finally, we utilized random forest modelling to identify the environmental factors most significantly associated with algal responses to the environmental-ecological controls (i.e., $\ln RR$ of environmental factors and phytoplankton relative abundances). Before the modelling, we calculated the correlations between different environmental factors. The results showed that there are no environmental factor pairs with a correlation greater than 0.8, which ensured that our analysis was not overfitting (Table S1). Model cross-validation used 75% of the dataset for training and 25% for validation at each resampling iteration (Xu & Goodacre, 2018). We used 500 trees for model training and sampled 2 predictors per node to minimize root-mean-square error (RMSE). After 1,000 simulations, we calculated the mean values of RMSE and variable importance across all models.

3 Results

3.1 Algal Density Reduction and Community Composition Changes after the Controlling

Through the one-year of environmental-ecological controlling, we observed a significant decrease in algal density across all three reservoirs (Fig. 2, Fig. S1 and Table S3). Overall, the average total phytoplankton density significantly decreased by 87.2%

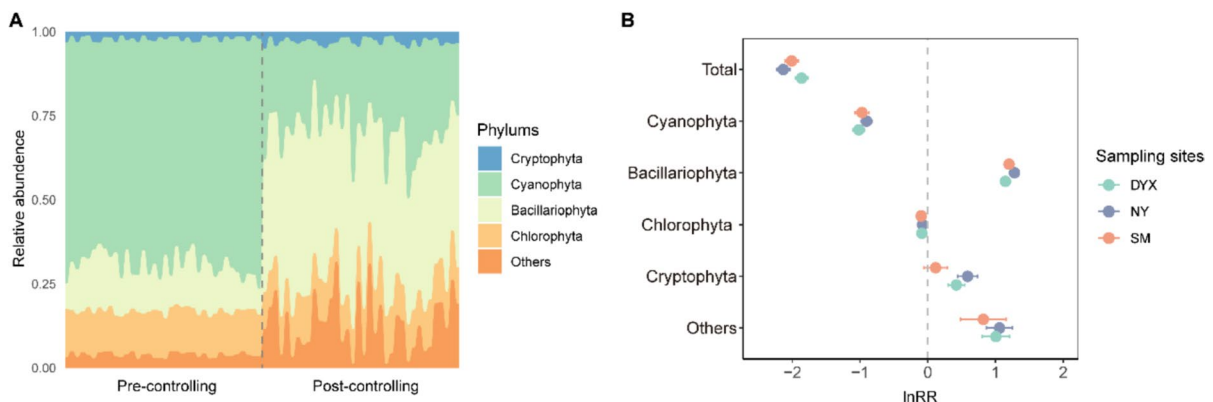


Fig. 2 Changes in phytoplankton community compositions and density pre- and post-controlling. **A.** The relative abundance of phytoplankton phyla, including Cyanophyta, Bacillariophyta, Chlorophyta, Cryptophyta, and others, before and after the ecological controlling. **B.** The natural log response Ratio ($\ln RR$) of Phytoplankton Phyla Density. This panel

shows the $\ln RR$ of total phytoplankton density and the relative abundance of individual phyla (Cyanophyta, Bacillariophyta, Chlorophyta, Cryptophyta, and others) for each sampling site (DYX: Dongyaxi, NY: Niuyang, SM: Shanmei). The positive $\ln RR$ values indicate an increase, while negative $\ln RR$ values indicate a decrease in density post-controlling

(Table S4), with reductions of 85.1% ($t=-19.6$), 88.7% ($t=-22.3$), and 87.3% ($t=-20.6$, $P<0.001$) in DYX, NY, and SM, respectively (Fig. 2b). Additionally, the structures of the phytoplankton community compositions also changed significantly during the intervention. The overall major detected phytoplankton phyla were *Chlorophyta*, *Bacillariophyta*, *Cryptophyta*, and *Cyanophyta*, and the dominant phyla were changed from *Cyanophyta* to *Bacillariophyta* after the controlling. The relative abundance of *Chlorophyta*, *Bacillariophyta*, and *Cryptophyta* significantly increased with restoration, while the relative abundance of *Cyanophyta* significantly decreased. Specifically, the relative abundance of *Cyanophyta* decreased by 62.5% ($t=-19.6$), 58.2% ($t=-11.2$), and 60.5% ($t=-9.6$) in DYX, NY, and SM, respectively, while the relative abundance of *Bacillariophyta* increased by 206.6% ($t=15.9$), 251.1% ($t=19.9$), and 223.6% ($t=15.9$) in DYX, NY, and SM, respectively ($P<0.001$; Fig. 2b). Additionally, the relative abundance of other minor phytoplankton species showed strong enhance, suggesting an overall increase in phytoplankton diversity after the control (Fig. 2, Table S3 and Table S4).

3.2 Changes in Environmental Factors and their Link to Algal Communities

Overall, algal density and community composition showed significant correlations with environmental factors (Fig. S2). Total phytoplankton density was significantly correlated with pH ($r=0.28$, $P<0.05$). Specifically, Spearman rank correlation results indicated that the relative abundances of *Cyanophyta* and *Chlorophyta* were generally consistent in relation to environmental factors, showing significant negative correlations with the DO and pH ($r=-0.25$ to -0.72 , $P<0.05$), and significant positive correlations with nitrogen and phosphorus content in sediments (such as TP and TN, $r=0.28$ to 0.78 , $P<0.05$). Additionally, the relative abundance of *Cyanophyta* was positively regulated by the permanganate index ($r=0.35$). As expected, *Cyanophyta* and *Chlorophyta* were identified as the primary contributors to increased Chla levels, with other minor taxa potentially exhibiting similar trends (Fig. S2). In contrast, the relative abundances of *Bacillariophyta* and *Cryptophyta* showed significant positive correlations with DO and pH ($r=0.23$ to 0.70), and significant negative

correlations with Chla, TN, TP, COD_{Mn} , and $\text{NH}_4^+\text{-N}$ contents in both water and sediments ($r=-0.30$ to -0.76 , $P<0.05$).

Furthermore, our results indicated that the environmental-ecological control measures significantly regulated the distribution of environmental factors and significantly alleviated eutrophication in water and sediments. One year after the implementation of the experiment, water pH, water Chla, water COD_{Mn} , water TP, water TN, water $\text{NH}_4^+\text{-N}$, sediment TP, sediment TN and sediment $\text{NH}_4^+\text{-N}$ were averagely decreased 15.4%, 35.8%, 49.1%, 73.5%, 73.4%, 41.1%, 69.1%, 22.1% and 42.2% in the three reservoirs ($t=-26.3$ to -4.1 , $P<0.05$), respectively, while the water DO significantly increased 81.9% ($t=8.9$ to 11.3 , $P<0.001$; Tables S5 and S6). The results also showed that the application of our experimental equipment was consistent in managing eutrophication in different reservoirs (Fig. 3). In addition, DO levels in treatment zones increased significantly from 3.2 ± 0.5 mg/L to 5.9 ± 0.5 mg/L in reference zones ($t=5.6$ to 6.8 , $P<0.001$), which confirms the

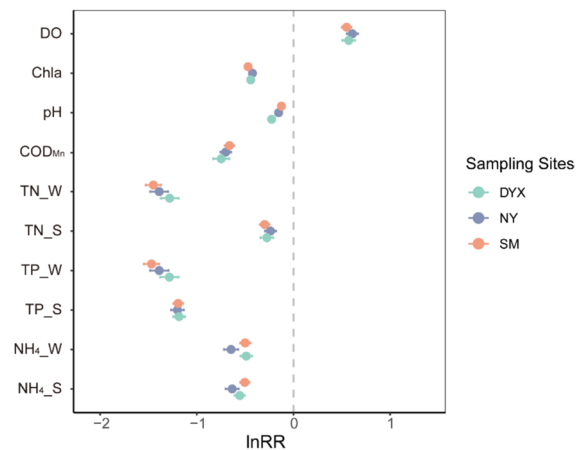


Fig. 3 Changes in environmental parameters pre- and post-controlling. The natural log response ratios (lnRR) of various environmental parameters measured in three reservoirs (DYX: Dongyaxi, NY: Niuyang, SM: Shanmei) before and after ecological controlling were shown. Positive lnRR values indicate an increase, while negative lnRR values indicate a decrease in density post-controlling. The error bars represent the 95% confidence intervals. The environmental parameters included water dissolved oxygen (DO), water chlorophyll-a (Chla), water KMnO_4 index (COD_{Mn}), water pH (pH), water total nitrogen (TN_W), water total phosphorus (TP_W), water ammonia nitrogen ($\text{NH}_4\text{-W}$), sediment TN (TN_S), sediment TP (TP_S), and sediment ammonia nitrogen ($\text{NH}_4\text{-S}$)

oxygenation of the nano aeration system can effectively cover the entire reservoirs.

3.3 Impact of Environmental-Ecological Controlling Measures and Predictive Modeling

The relative abundance of phytoplankton, particularly the *Cyanophyta*, was closely linked to changes in environmental factors in response to the environmental-ecological control measures (Fig. 4). The linear mixed-effects models (LMMs) indicated that, even when accounting for the heterogeneity among reservoirs, the increases in nitrogen, phosphorus, and ammonium concentrations significantly positively impacted the relative abundance of *Cyanophyta* (with exception, sediment ammonium nitrogen concentration). This suggests that, even after ecological controlling, eutrophication can still potentially lead to algal blooms. Therefore, long-term water quality monitoring is essential, including continuous aeration to increase dissolved oxygen levels. In addition, the lnRR of total phytoplankton density was significantly negatively correlated with the lnRR of COD_{Mn} , water TN and TP, while other phylum showed significant

positive correlations with changes in DO and sediment TN (Fig. S3).

The random forest models further supported previous findings (Fig. 5), indicating that combining environmental and ecological control measures are effective in reducing environmental nutrient levels, increasing dissolved oxygen, thus reducing phytoplankton density, and preventing algal blooms. Specifically, changes in COD_{Mn} , TP, and DO were the strongest predictors of changes in total phytoplankton density, as well as the relative abundance of *Cyanophyta* and *Bacillariophyta*, respectively. Additionally, changes in Chla levels may further significantly influence the relative abundances of *Chlorophyta* and other phyla.

4 Discussions

By applying nano-aeration and environmentally friendly microbiological techniques to three typical eutrophic reservoirs in Fujian Province, we demonstrated significantly reduced density and changed the compositions of phytoplankton community (Figs. S1

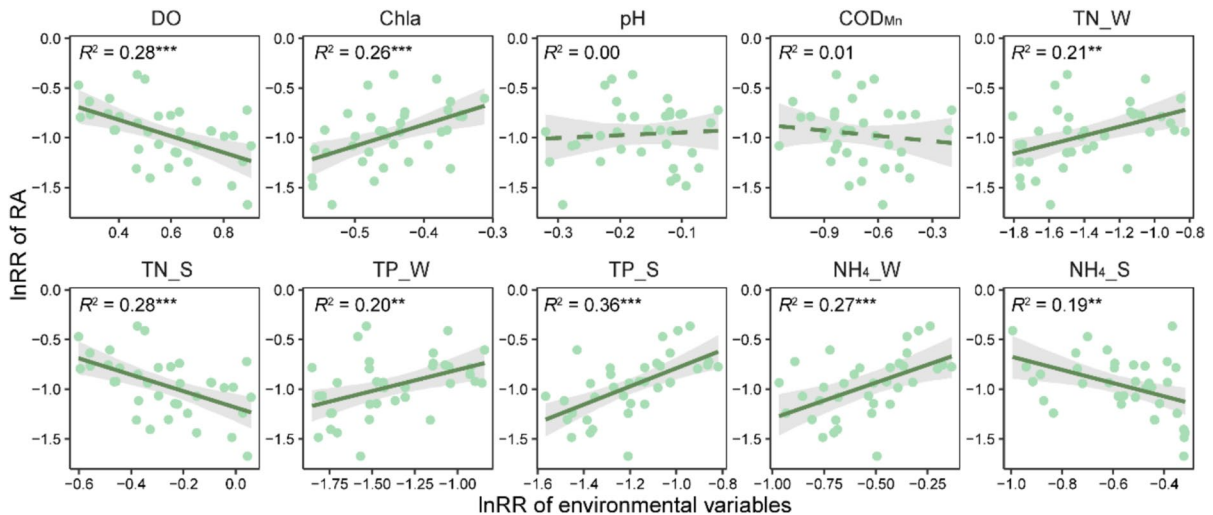


Fig. 4 Correlations between the natural log response ratio (lnRR) of relative abundance (RA) of Cyanophyta and the lnRR of various environmental variables. Each subplot includes the coefficient of determination (R^2) indicating the strength and significance of the correlation. The regression lines indicate the direction and magnitude of these relationships, extracted from the linear mixed-effects models (see section of Materials and Methods). The grey areas around

the regression lines represent the 95% confidence intervals. The environmental parameters included water dissolved oxygen (DO), water chlorophyll-a (Chla), water KMnO4 index (COD_{Mn}), water pH (pH), water total nitrogen (TN_W), water total phosphorus (TP_W), water ammonia nitrogen ($\text{NH}_4\text{-W}$), sediment TN (TN_S), sediment TP (TP_S), and sediment ammonia nitrogen ($\text{NH}_4\text{-S}$)

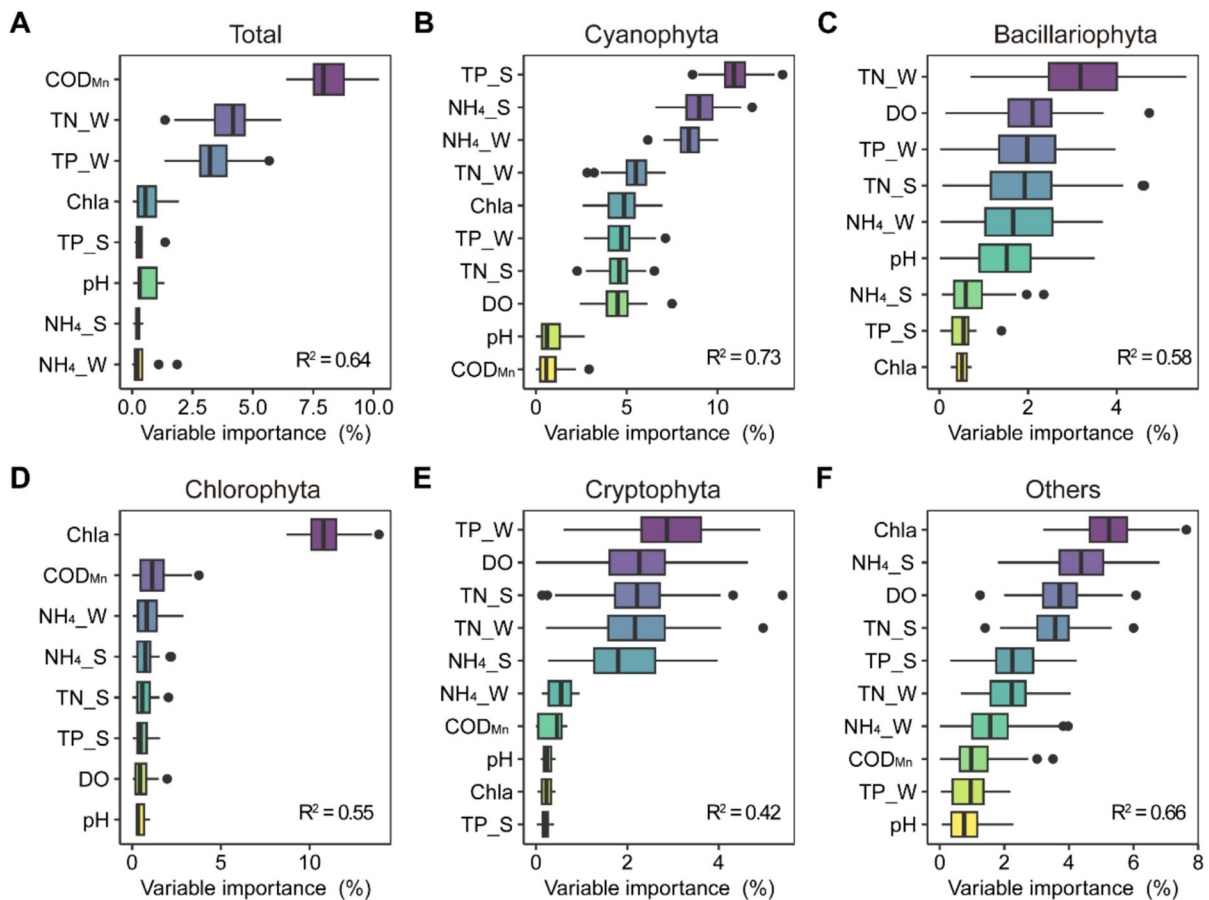


Fig. 5 Variable importance plots from random forest models. We used the models to predict changes in the total phytoplankton density and the relative abundances of different phytoplankton phyla (Cyanophyta, Bacillariophyta, Chlorophyta, Cryptophyta, and other phylum) by environmental variables when post-controlling. Each plot highlights the key predictors for each group. Only the variables with positive importance

were shown. The environmental parameters included water dissolved oxygen (DO), water chlorophyll-a (Chla), water KMnO₄ index (COD_{Mn}), water pH (pH), water total nitrogen (TN_W), water total phosphorus (TP_W), water ammonia nitrogen (NH₄_W), sediment TN (TN_S), sediment TP (TP_S), and sediment ammonia nitrogen (NH₄_S)

and 2). The substantial reduction in total phytoplankton density by 87.2% and in relative abundance of *Cyanophyta* by 60.3% suggests the effectiveness of our controlling methods in reducing HABs' proliferation. Phytoplankton blooms in our study were mainly caused by the specific phyla of *Cyanophyta*, which thrives in conditions of 25 and 30 °C, high alkalinity, sufficient nitrogen and phosphorus, low currents, and full sunlight (Thajuddin & Subramanian, 2005). Their gas vesicle-mediated buoyancy regulation gives them a growth advantage in such environments, especially in reservoirs' stagnant zones that promote nutrient accumulation (Visser et al., 2016). Notably, our combined technique outperforms single

hydrodynamic interventions (e.g., Song et al. (Song et al., 2023), which achieved a 36% reduction in biomass of *Cyanophyta* via reservoir operation optimization) by directly disrupting both nutrient availability and cyanobacterial niche adaptation, highlighting the synergistic benefits of physical aeration and biological nutrient cycling.

The results from this study are consistent with previous findings where nano-aeration and microbial technologies have been successfully applied to manage eutrophic lakes and urban rivers. For instance, in urban black-odor rivers, nano-aeration technologies significantly enhance dissolved oxygen levels, reduce nutrient concentrations compared to conventional

aeration methods (Xu et al., 2023). Simultaneously, environmentally friendly microbiological techniques have gained attention as sustainable solutions for eutrophication management (El-Sheekh et al., 2021). Microorganisms such as *Actinobacteria*, *Firmicutes*, and *Myxococcota* play a crucial role in nutrient cycling and organic matter degradation (Kakade et al., 2021). These microbial communities can enhance the self-purification capacity of water bodies through processes such as denitrification, phosphorus sequestration, and degradation of harmful *Cyanophyta* (Zhang et al., 2020). For instance, specific bacterial strains like *Pseudomonas* and *Bacillus* have been shown to degrade organic matter, reduce total nitrogen and phosphorus, and suppress growth of *Cyanophyta* (Bi et al., 2019).

Our results also showed that environmental-ecological control measures can reduced the eutrophication levels of small water bodies (such as reservoirs) in a relatively short period of time (Fig. 3 and Table S3). The interactions between environmental factors and eutrophication are complex and multifaceted. Eutrophication frequently arises from nutrient pollution (predominantly phosphorus and nitrogen), which results in increased primary production and algal blooms (Glibert, 2017). In freshwater systems, phosphorus is typically nutrient that is limiting, and an increase in the amount of phosphorus can have a significant effect on algal growth (Carpenter, 2008). It is evident that elevated nutrient levels, in conjunction with warm temperatures, engender optimal conditions for the proliferation of HABs. Thus, managing the environmental factors could be essential for controlling eutrophication and preventing HABs.

The environmental-ecological control measures (i.e., nano-aeration and microbial intervention) significantly affected other environmental factors such as water pH and COD_{Mn} that correlated with the algal community dynamics. The reduction of pH has been demonstrated to inhibit phytoplankton species that are able to propagate readily under alkaline conditions, such as *Cyanophyta* (Wang et al., 2024a). Moreover, the reduction in COD_{Mn} indicates a significant reduction in organic pollutants in the water, which in turn reduces the risk of water quality deterioration and promotes the restoration of the water ecosystem (Shi et al., 2024). It is hypothesized that these enhancements may also serve as the basis for the restoration of underwater vegetation and the stabilization of the

ecosystem (Araya-López et al., 2024). Moreover, the efficacy of these control measures has been demonstrated by their consistent performance across the three reservoirs, signifying their wide applicability and significant promotion value.

The relative abundances of *Cyanophyta* could be most predictable by environmental variables (such as TP and TN) during the environmental-ecological control (Figs. 4 and S3). Environmental factors, particularly nutrient levels (such as concentrations of total phosphorus and total nitrogen) and dissolved oxygen, play a crucial role in the dynamics of algal blooms. Our results showed that the relative abundances of *Cyanophyta* were significantly correlated with nutrient levels, particularly for TP and TN. After the controlling, *Bacillariophyta* became the most dominant phylum, which is significant for several reasons (Fig. 2). *Bacillariophyta* are known for their role in enhancing water quality due to their silica-based cell walls, which are less prone to decomposition and do not contribute to nutrient recycling in the same way as *Cyanophyta* (Bhat et al., 2025). Thus, the increase in *Bacillariophyta* suggests an improvement in water quality and a shift towards a more stable and less harmful phytoplankton community (Desrosiers et al., 2013). *Bacillariophyta* thrives in well-oxygenated and nutrient-balanced environments, indicating that the controlling efforts successfully created conditions unfavorable for harmful algal blooms. Importantly, our results highlight that nutrient levels continue to play a dominant role in driving *Cyanophyta* dynamics even after the ecological control measures (Fig. 5). This indicates the necessity of long-term monitoring and adaptive management strategies to ensure the sustained success of eutrophication management efforts.

5 Conclusion

In conclusion, our study emphasizes the necessity of ongoing monitoring and adaptive management to sustain the improvements achieved through ecological controlling. The integration of nano-aeration and environmentally friendly microbial techniques offers a promising solution to the challenges of eutrophication and harmful algal blooms, providing a sustainable path forward for water quality management. Understanding these dynamics is crucial

for developing effective, sustainable strategies for managing and preventing harmful algal blooms in the future.

Several limitations should be acknowledged. First, the observed improvements may be confounded by external factors (e.g., reduced anthropogenic nutrient inputs and climate variability) (Priya et al., 2023). Second, nano-aeration-induced water disturbance could potentially mobilize heavy metals (e.g., lead, cadmium) from anoxic sediments into the water column (Zahir et al., 2024). Third, the efficacy of the integrated techniques may vary across environmental settings, as oligotrophic reservoirs with low nutrient reserves or highly stratified deep reservoirs may show weaker phytoplankton community shifts (Wang et al., 2024b). Fourth, the inoculation of functional microorganisms may exert unintended impacts on native microbial communities, such as competitive exclusion of indigenous nutrient-cycling bacteria or disruption of trophic cascades (Sheng et al., 2025). Future research should address these gaps by: (1) quantifying metal mobilization risks via sediment core incubation experiments; (2) testing the techniques in diverse reservoir types (e.g., oligotrophic, deep stratified) to clarify environmental suitability; (3) using high-throughput sequencing to track changes in native microbial community structure post-inoculation; and (4) incorporating vertical sampling and continuous environmental monitoring to disentangle the effects of interventions from natural variability.

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Authors Contributions CL conceived the idea. PL conducted the research, developed the methods and interpreted the data. PL and CL performed the data analysis. PL prepared the figures and tables. PL, CL, QS, ML, ZG, and SH wrote the manuscript.

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Data Availability The data presented in this study are available upon request from the corresponding author. Due to the ongoing project and pending patent applications for functional

microbial agents, the 16S rRNA gene sequences of isolated strains are not publicly available currently. Original sequencing data and phylogenetic tree files related to bacterial identification will be made available to interested researchers upon reasonable academic request.

Declarations

Ethics Approval Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no conflict of interest.

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